5.Experiment: Apache Spark Basics – RDDs and DataFrames

AIM

To understand and implement basic operations in **Apache Spark**, including:

- Creation of **RDDs (Resilient Distributed Datasets)** and **DataFrames**
- Performing **transformations** and **actions** on them

DESCRIPTION

Apache Spark is a fast, in-memory distributed computing framework used for big data processing.

Key Concepts:

1. RDD (Resilient Distributed Dataset)

- Fundamental data structure in Spark
- Immutable, distributed collection of objects
- Supports parallel processing

2. DataFrame

- Distributed collection of data organized into named columns
- Similar to a table in relational databases
- Optimized using Catalyst optimizer

3. Transformations

- Operations that create a new dataset from an existing one
- Lazy in nature (not executed immediately)
- Examples:
    - map()
    - filter()
    - flatMap()

4. Actions

- Operations that return results or write data
- Trigger execution of transformations
- Examples:

- collect()
- count()
- show()

STEP-BY-STEP EXPLANATION

Step 1: Install and Setup Spark

- Install Spark and Python
- Start PySpark shell or use Jupyter Notebook

Step 2: Create Spark Session

- Entry point to Spark functionality

Step 3: RDD Creation

- Create RDD from a list

Step 4: Apply Transformations on RDD

- Use map() and filter()

Step 5: Apply Actions on RDD

- Use collect() and count()

Step 6: Create DataFrame

- Convert list or RDD into DataFrame

Step 7: Apply Transformations on DataFrame

- Select columns
- Filter rows

Step 8: Perform Actions on DataFrame

- Display results using show()

SOURCE CODE (PySpark)

```
# Step 1: Import SparkSession
from pyspark.sql import SparkSession

# Step 2: Create Spark Session
spark = SparkSession.builder \
    .appName("Spark Basics") \
    .getOrCreate()

# Step 3: Create RDD
data = [1, 2, 3, 4, 5]
rdd = spark.sparkContext.parallelize(data)

# Step 4: Transformations on RDD
rdd_map = rdd.map(lambda x: x * 2)
rdd_filter = rdd.filter(lambda x: x % 2 == 0)

# Step 5: Actions on RDD
print("Mapped RDD:", rdd_map.collect())
print("Filtered RDD:", rdd_filter.collect())
print("Count:", rdd.count())

# Step 6: Create DataFrame
data_df = [("Alice", 25), ("Bob", 30), ("Cathy", 22)]
df = spark.createDataFrame(data_df, ["Name", "Age"])

# Step 7: Transformations on DataFrame
df_filter = df.filter(df.Age > 23)
df_select = df.select("Name")

# Step 8: Actions on DataFrame
print("Original DataFrame:")
df.show()

print("Filtered DataFrame:")
df_filter.show()

print("Selected Column:")
df_select.show()
```

```
# Stop Spark Session
spark.stop()
```

OUTPUT

```
Mapped RDD: [2, 4, 6, 8, 10]
Filtered RDD: [2, 4]
Count: 5

Original DataFrame:
+-----+---+
| Name|Age|
+-----+---+
|Alice| 25|
|  Bob| 30|
|Cathy| 22|
+-----+---+

Filtered DataFrame:
+-----+---+
| Name|Age|
+-----+---+
|Alice| 25|
|  Bob| 30|
+-----+---+

Selected Column:
+-----+
| Name|
+-----+
|Alice|
|  Bob|
|Cathy|
+-----+
```

RESULT

Successfully implemented:

- Creation and manipulation of **RDDs**
- Creation and querying of **DataFrames**
- Execution of **transformations and actions** in Spark

Thus, the basic operations of **Apache Spark** were learned and verified.